

EXP 5

Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

Code :

```
import numpy as np
```

```
X = np.array([
    [2, 60],
    [3, 65],
    [5, 75],
    [6, 80],
    [8, 90]
])
```

```
y = np.array(["Fail", "Fail", "Pass", "Pass", "Pass"])
```

```
test = np.array([7, 85])
```

```
k = 3
```

```
distances = np.sqrt(np.sum((X - test) ** 2, axis=1))
```

```
k_indices = np.argsort(distances)[:k]
```

```
k_nearest_labels = y[k_indices]
```

```
prediction = max(set(k_nearest_labels), key=list(k_nearest_labels).count)
```

```
print("Training Data:")
```

```
print(X)
```

```
print("\nClass Labels:")
```

```
print(y)
```

```
print("\nTest Sample:", test)
```

```
print("\nDistances:")
```

```
print(distances)
```

```
print("\nNearest Neighbours:")
```

```
print(k_nearest_labels)
```

```
print("\nPredicted Class:", prediction)
```

Output

Python 3.14.6

File Edit Shell Debug Options Window Help

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

= RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/machine learning lab/exp 5.py

Training Data:

```
[[ 2 60]
 [ 3 65]
 [ 5 75]
 [ 6 80]
 [ 8 90]]
```

Class Labels:

```
['Fail' 'Fail' 'Pass' 'Pass' 'Pass']
```

Test Sample: [7 85]

Distances:

```
[25.49509757 20.39607805 10.19803903  5.09901951  5.09901951]
```

Nearest Neighbours:

```
['Pass' 'Pass' 'Pass']
```

Predicted Class: Pass

>>>